

QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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QUALITY ASSURANCE & TESTING DOCUMENTATION (QATD)

Project Name: BizMind – AI Business Advisor

Version: 1.0

1. Introduction

This section introduces the testing strategy used to validate the **Decision Intelligence** accuracy of BizMind.

1.1 Purpose

The purpose of this document is to provide a reference for developers and judges to ensure the high-level architecture handles data correctly. It highlights the technical scope and the testing-related decisions behind the BizMind development.

1.2 Testing Strategy

We focus on validating the "Logic-First" pipeline:

- **Math Accuracy:** Ensuring the Python backend calculates variances correctly before sending them to the AI.
- **Response Latency:** Ensuring the Z AI GLM returns a response within 3–5 seconds to maintain user experience.

- **Prompt Integrity:** Validating that the context window is properly constructed with factual data.

2. Test Plan & Scope

2.1 Test Environments

Type	Details
Backend Testing	Python pytest for compute_sales_metrics logic.
API Testing	Validating JSON request/response formats between FastAPI and Z AI GLM.
LLM Testing	Prompt evaluation to ensure no "hallucinations" in business recommendations.

2.2 Stakeholder Expectations (Testing Focus)

Stakeholder	Roles	Expectations
Customer (Adam)	Enters business data	Receives accurate, non-gibberish business insights.
Development Team	Maintain logic/AI sync	Clear API usage contracts and modular code.
QA Team	Validate system behaviour	Coverage of edge cases (e.g., zero sales days).

3. Test Cases (Functional Requirements)

ID	Feature	Test Case Description	Expected Result	Actual Results (Evidence)	Status
TC-01	Sales Logic	Enter Weekday RM200 and Weekend RM500.	Backend computes exactly +150% growth.	{"growth": 150.0, "status": "success"}	Passed
TC-02	AI Reasoning	Feed "150% growth" fact to Z AI GLM.	AI recommends strengthening weekday revenue.	"1. Focus on improving weekday traffic with promotions..."	Passed
TC-03	Contextual Prompt Injection	Verify that the RM500 weekend sales figure is correctly passed into the AI's reasoning context.	The AI response must explicitly reference the RM500 figure in its recommendation.	AI response included: "...since your RM500 weekend revenue is high, consider..."	Passed
TC-04	Zero-Value Input	Enter RM0 on weekends to simulate a day off.	System should handle the math without a "Division by Zero" error.	Displayed: "0% growth detected. Strategy: Optimize costs."	Passed

4. Non-Functional Requirements (NFR) Validation

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Quality	Requirement	Implementation Result
Token Latency	API calls must return within 3s.	Passed: Measured average response time of approx. 1.5s using asynchronous httpx calls.
Reliability	Math must be 100% accurate.	Passed: Python Decimal library used for RM values; zero rounding errors in 20/20 test runs.
Cost Efficiency	Fixed prompt lengths.	Passed: Prompt tokens limited to 500 per request to minimize API costs per user session.

5. Summary & Evaluation

5.1 Emergency Rollback Strategy

If anomalies are detected in the AI reasoning (e.g., gibberish output), the system triggers an Emergency Rollback to the **Golden Version**, which relies on hard-coded business rules until the API is stable.

5.2 Assumptions

- Users have a stable internet connection during data submission.
- The Z AI GLM API provides consistent JSON structured output.